

A Real Time Facial Emotion Recognition Using Depth Sensor And Interfacing With Second Life Based Virtual 3D Avatar

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Abstract—Facial emotions are the nonverbal form of communication and we find it very useful when we have no words to express our feelings then we use the gestures to express the same. To detect and track emotions dynamically, we have to use Kinect Sensor since it has camera to produce 3D depth maps. The Facial emotions are detected in real-time, by applying the mesh over the tracked face and thereby identifying the desired points for extracting the features. Facial Action Code Systems (FACS) and Facial Animated Parameters (FAP) are the places of interest for depicting the emotions.

Second Life is the 3D Virtual world where people are able to create a digital character called 'Avatar' and thereby interacting with the people in the virtual world. The Avatar, which shows some gestures, needs some emotions in a particular scenario. Using the real-time facial emotion recognition based on Kinect depth sensor, avatar emotions are also generated in the 3D Virtual world using Second Life. This idea can be further extended to serve as a communication link so that speech and hearing impaired people can express their emotions.

Keywords—Depth sensor; Second Life; Avatar; Facial Action Coding System (FACS); Action Units (AU)

I. INTRODUCTION

Kinect was first launched at 2010 and Microsoft had launched Kinect for Windows operating system in 2012. The name of Kinect comes from the word “kinetic” and “connection”. Kinect Sensor is very popular all over the world and becomes the fastest selling consumer electronics device for playing video games. Kinect is motion sensing input device developed by Microsoft, and mainly used for the equipment of Xbox 360 and Windows. They could interact and control the Xbox 360 by their gestures and movement of body without using the remote control. Developers could study Kinect SDK and develop program in C++, C# or Visual Basic.

The Natural User Interface (NUI) is the core of the Kinect for Windows API. Through the NUI, developer would get the sensor data such as audio, depth and image stream in application. There are two methods for getting the image frames, polling and event models. The polling model is used to read data frames. The event model supports the ability to

use those data streams with more accuracy and flexibility. The Architecture is shown below in “Fig.1”

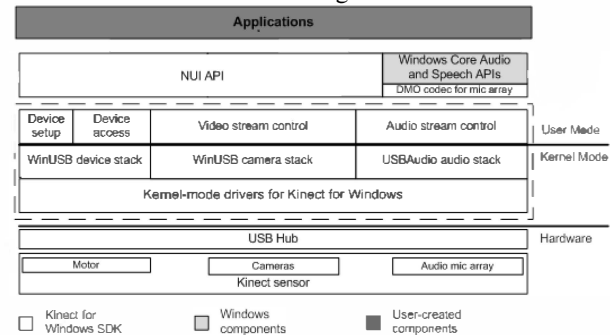


Fig. 1. Architecture of Kinect depth sensor

Second life is celebrating its 10th anniversary where users plan to have parties, live music, DJ events and celebrations. Second life is an online 3D virtual world. It took the shape in the Linden Lab. It provides an immersive environment where users are able to create a digital character or 'avatar', and interact with people from around the world. Second life allows users or 'residents' to come together to engage in a range of activities including socializing, attending musical events, playing games, shopping, running business, building, collaborating, attending meetings and presentations and undertaking training and education.

The most interest in Face Detection of Kinect SDK is to achieve project, Emotion Recognition. For the face tracking, there are many features. To start with, the face tracking includes tracking the face position, orientation and the face features in real time. Moreover, a 3D mesh of face, including the mouth and eyebrow is animated. Last but not least, it would track several faces at the same time.

Face tracking SDK is used for the developer to develop application about face detection. The face tracking SDK engine analyzes the input from the Kinect camera. It then estimates the head pose and facial expressions. Finally, the information would drive NUI or used in other application. Technical Specifications Input image Face Tracking SDK allows Kinect depth and color images as input. The tracking quality would be affected by the image quality of input frames. For example the darker condition would be harder for

tracking the face. Surprisingly, larger faces are easily tracked than smaller faces.

Face tracking SDK use Kinect coordinate system to output 3D tracking results. The camera's optical center is the origin. Y axis is pointing up while Z axis is pointing towards the user. The measurement units are meters for translation while the measurement units are degrees for rotation angles. Finally, the computed 3D mesh has coordinates which place it over the player's face can be shown in "Fig.2"

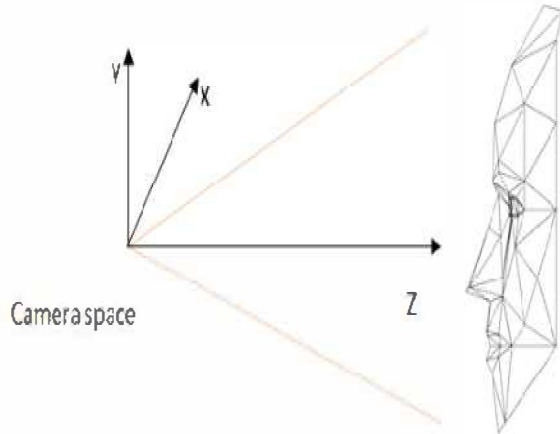


Fig. 2. Face tracking in 3D coordinate system

Face Tracking Outputs - The output of the Face Tracking engine contains the following information about a tracked user:

- Tracking status
- 2D points
- 3D head pose
- Facial Mesh

II. LITERATURE SURVEY

A. Background

Accurate and automatic tracking of people in the virtual world [1] is an interesting problem for terrorism and security experts. One way to track and recognize the identity of a person in the virtual world is to recognize the face of the avatar that represents this person. To deal with this problem there are two algorithms capable of recognizing avatar faces with a high degree of accuracy. Experiments conducted on two datasets of avatars from the Second Life and Entropies Universe virtual worlds show the effectiveness of using these algorithms for recognizing avatar faces. Paul Ekman [2], well known psychologist has done extensive research on universal, basic expressions that are known within all cultures, resulting in a list of the following six basic emotions: Smile, Sad, Surprise, Fear, and Anger. The emotion of contempt was added to the list by the authors of the Cohn-Kanade database [7].

The emotion recognition system based on facial expression is described by fully automatic facial expression recognition

system [3] is based on three steps: face detection, facial characteristic extraction and facial expression classification. An anthropometric model is developed to detect facial feature points. The variations of 21 distances which describe the facial features deformations from the neutral face were used to coding the facial expression. Classification step is based on SVM method (Support Vector Machine). Experimental results demonstrate that it is an effective method to recognize emotions through facial expression with an emotion recognition rate more than 90% in real time. This approach is used to control music player based on the variation of the emotional state of the computer user.

Creation of Emotion Avatar Image (EAI) [4] as a single good representation for each video or image sequence for emotion recognition analyzes the spatial and temporal information for every single frame in a human emotion video. SIFT flow algorithm can be introduced to register every frame with respect to avatar reference face model. Extracting the features from EAIs using both Local Binary Pattern (LBP) and Local Phase Quantization (LPQ) the results of the textures are tested on the Facial Expression Recognition and Analysis Challenge (FERA-2011) data, GEMEP-FERA dataset [8].

The efficiency of the approach for biologically [5] inspired fusion that emulates the process of recognition facial emotion from video is 18.5 % on publicly available data. In this both humans and computer underperform. In recent emotion recognition competitions, two approaches were using either geometric relationships that best captured facial dynamics or an accurate registration technique to develop appearance features. These two methods capture two different types of facial information similarly to how the human visual system divides information when perceiving faces.

The problem of Multimedia Emotion Recognition System (MERS) [6] was dealt by non-verbal upper body expression gestures. System tries to compute a probability of the emotional state of a user using simple multimedia peripherals. All aspects of human computer interaction in the multimedia frameworks, try to be adapted with all feedbacks that can be detected from users. This adaptation can take the name of a social dimension.

B. Outcome Of Literature Survey

After a thorough study of the recent research papers published in the field of facial emotion detection, a generic method was selected that was implementable and modifiable. Six basic human emotions were selected to be recognized by the Kinect Depth Sensor using smile, surprise, sad, anger, fear and disgust emotions.

These are the step by step procedure identified to implement the system

1. Detecting the face using Kinect sensor
2. Detecting emotions using Action Units (AU) of Facial Action Coding System (FACS)
3. Creating emotions to avatar in 3D Virtual world of Second Life

4. Interfacing the emotions tracked by Kinect depth sensor to 3D Avatar of Second Life.

C. Problem Statement

To design an efficient Facial Emotion Detection system using Depth Sensor in Kinect and interfacing the emotions to a virtually created 3D avatar of the Second Life.

D. Objectives

These are the objectives of the whole project

1. To create facial mesh over the tracked face.
2. To locate the points that are playing major role in emotion detection.
3. To analyze and detect the emotion
4. Creating emotions to avatar of Second Life.
5. To interface the emotions detected by Kinect sensor with Second Life.

III. METHODOLOGY AND FRAMEWORK

The Facial Action Coding System (FACS) Manual is a detailed, technical guide that explains how to categorize facial behaviors based on the muscles that produce them, i.e., how muscular action is related to facial appearances. It illustrates appearance changes of the face using written descriptions, still images, and digital video examples. Behavioral scientists, CG animators, computer scientists interested in pattern recognition programs, and other technicians and scientists use FACS in their professional work when they need to know the exact movements that the face can perform, and what muscles produce them. Working through the exercises of the FACS Manual may also enable greater awareness of and sensitivity to subtle facial behaviors that could be useful for psychotherapists, interviewers, and other practitioners who must penetrate deeply into interpersonal communications.

Facial Action Coding System (FACS) is the most widely used and versatile method for measuring and describing facial behaviors. Paul Ekman and W.V. Friesen developed the original FACS in the 1970s by determining how the contraction of each facial muscle (singly and in combination with other muscles) changes the appearance of the face. They examined videotapes of facial behavior to identify the specific changes that occurred with muscular contractions and how best to differentiate one from another. They associated the appearance changes with the action of muscles that produced them by studying anatomy, reproducing the appearances, and palpating their faces. Their goal was to create a reliable means for skilled human scorers to determine the category or categories in which to fit each facial behavior. The FACS Manual was first published in a loose-leaf version with video or film supplements in 1978.

FACS measurement units are Action Units (AUs), not muscles, for two reasons. First, for a few appearances, more

than one muscle was combined into a single AU because the changes in appearance they produced could not be distinguished. Second, the appearance changes produced by one muscle were sometimes separated into two or more AUs to represent relatively independent actions of different parts of the muscle. (After all, facial muscles were identified and named by anatomists, not behavioral psychologists.)

A FACS coder "dissects" an observed expression, decomposing it into the specific AUs that produced the movement. The scores for a facial expression consist of the list of AUs that produced it. Duration, intensity, and asymmetry can also be recorded.

A. Implementation Of Emotions In Real World Using Kinect

Emotions can be tracked by various methods; the more effective one is by using mesh. The Action Units (AUs) of the Facial Action Coding System (FACS) plays a vital role in sketching the change of the emotions of the detected face.

In this project, I identified the points which actually changes when the person shows his/her emotions. These points were taken under consideration, and readings were taken. The reading gave a slight variation in the distance. When the face is about 100cms away from the sensor, the readings were taken and the emotion tracking is achieved.

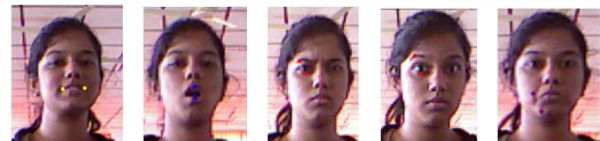


Fig. 3. Emotion tracking for smile, surprise, sad, fear and anger using kinect depth sensor

Primarily, the points which paved the way for the emotions are noted down. Later the readings are taken when the person shows the emotions. This calibrates the emotions over a specific threshold.

The following chart shows the flow of work done in detecting the emotions using Kinect sensor.

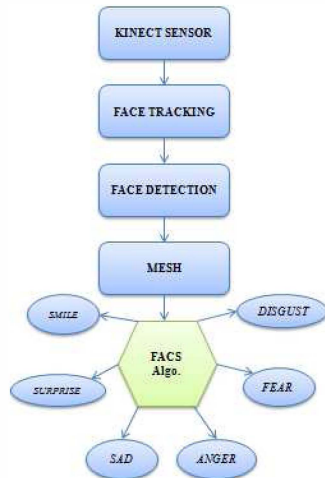


Fig. 4. Flow chart for generating emotions using Kinect sensor

“Fig. 4” shows the flow chart that represents the emotion generation using FACS Algorithm and implemented in real time using Kinect depth sensor. When the connection with Kinect depth sensor is established then it tracks for the human face. The next step is, it detects the face and later it applies facial mesh over the detected faces. It can detect more than one face at a time. Now, when our code of program is applied, it gives the result of the emotion carried out by the person in front of the Kinect Sensor.

B. Implementation Of Emotions In 3D Virtual World

The Second Life is facilitated with some gestures. These can be designed to express the avatar’s emotions to the other people in the 3D Virtual world.

Initially, the basic emotions of smile, surprise, sad, anger, fear and disgust are assigned to the avatar then achieved the emotions of the 3D Avatars in the Second Life as shown in “Fig. 5”.



Fig. 5. Emotion tracking for smile, surprise, sad, fear and anger using avatar in second life.

At the final stage, we adopted the emotions which were recognized in real – time from Depth sensor to the Avatar which is created in the 3D Virtual world of second life. To achieve this, we manipulated our previous code, which send keys to the Second Life. An extra class file is included which handles the program to send the keyboard events to another application. Here, the expressions obtained from the Kinect

sensor are to be connected to an avatar of second life.

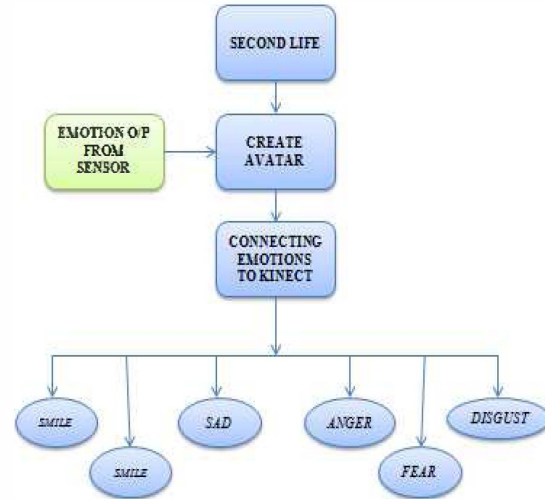


Fig. 6. Flow chart for generating emotions in second life based virtual 3D avatar.

“Fig. 6” shows the flow chart that represents the process of emotion display in Second life using 3D Avatars. Primarily, the Second Life is loaded. The avatar is created which possess the emotional gestures. Now, the Second Life is interfaced with the outputs of the Emotion detection. When the both Emotion detection and Second life are connected, it gives the emotion of the user, who is playing the emotion sitting.

IV. EMOTION DETECTION USING KINECT

FACS Algorithm based Real time Emotion Detection is implemented using Kinect Depth Sensor for 5 different emotions namely Smile, Surprise, Sad, Fear and Anger. The precision and efficiency of the Emotion Recognition is calculated. We took 10 sample expressions of which, each sample was calculated for 3 readings.

A. Smile

When we smile, our lips get widened. We can use this concept to detect the feature of smile. Generally the length of the lips is calculated to know the original length. Later, under the circumstances, that, when the person smiles, then the distance between lips increases. Thus, we can detect the smiling feature as shown in “Fig. 7”

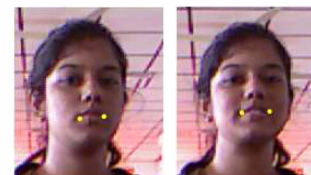


Fig. 7. Emotion – Smile

The points 88,89 are used to calculate the distance between edges of the lips as they are the two extreme points on either side respectively.

Later we calculated threshold where the entire project can work under the given circumstances (see Limitations). It is found that when a person smiles, the width of the mouth is greater than threshold. When we coded for the smile detection, it is,

```
if (mouthWidth > threshold)
{
    tb1.Text = "smile";
    //Address the smile of the avatar with the key stroke F2.
    keybd_event((byte)Keys.F2, 0x45, KEYEVENTF_EXTENDEDKEY, (UIntPtr)0);
    //This triggers the keyboard event to smile when the user smiles.
    this.ReleaseMouseCapture();
}
```

Fig. 8. Code snippet for smile.

A confusion matrix is used to describe the True positive, True Negative, False Positive and False negative of the system. In this paper, the following confusion matrices describes the same regarding the expressions calculated using the sensor.

TABLE I. CONFUSION MATRIX FOR SMILE

	Positive	Negative
True	28	2
False	0	30

B. Surprise

When we are surprise, there will be calculate-able vertical distance between upper and lower lip. So, under the given circumstances, when the person is surprised, there will be vertical distance between upper and lower lip. Thus, we can detect the surprise feature as shown in "Fig. 9"

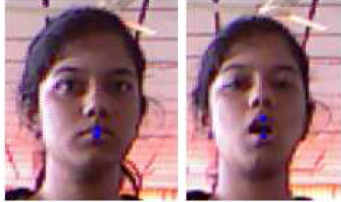


Fig. 9. Emotion – Surprise.

The points 7, 40 are used to calculate the distance between top and bottom edges of the lips.

We found that when a person is surprised, the vertical width of the mouth is greater than threshold. When we coded for the surprise detection, it is,

```
else if (openWidth > th)
{
    tb1.Text = "surprise";
    //Address the surprise of the avatar with the key stroke F3.
    keybd_event((byte)Keys.F3, 0x45, KEYEVENTF_EXTENDEDKEY, (UIntPtr)0);
    //This triggers the keyboard event to surprise when the user surprises.
}
```

Fig. 10. Code snippet for surprise

A confusion matrix is used to describe the True positive, True Negative, False Positive and False negative of the system. In this paper, the following confusion matrices

describes the same regarding the expressions calculated using the sensor.

TABLE II. CONFUSION MATRIX FOR SURPRISE

	Positive	Negative
True	29	1
False	1	29

C. Sad

When we are sad, there will be reduction in distance from lip edge to chin. This can be taken into consideration. So, under given circumstances, when the person is sad, there will be reduction in distance between chin and the edge point of lip. Thus, we can detect sad feature as shown in "Fig. 11"



Fig. 11. Emotion – Sad.

The points 32, 90 are used to calculate the distance between edge of the lip and chin.

We found that when a person is sad, the distance between chin and extreme edge of the lips is negligible. So we squared it to have a vast difference between values and found that it worked. When we coded for the sad detection, it is,

```
else if ((lineWidth * lineWidth) < 415)
{
    tb.Text = "sad";
    //Address the sadness of the avatar with the key stroke F4.
    keybd_event((byte)Keys.F4, 0x45, KEYEVENTF_EXTENDEDKEY, (UIntPtr)0);
    //This triggers the keyboard event to sad when the user is sad.
}
```

Fig. 12. Code snippet for sad

TABLE III. CONFUSION MATRIX FOR SAD

	Positive	Negative
True	25	5
False	0	30

D. Fear

When we are afraid, we raise our eyebrows. Which means the distance between your eyebrow and eye will be considerably large when compared to normal expression. So, under the given circumstances, when the person is afraid, there will be calculate-able difference of distance between eyebrow and eye. Thus, we can detect the feature of fear as shown in "Fig. 13"

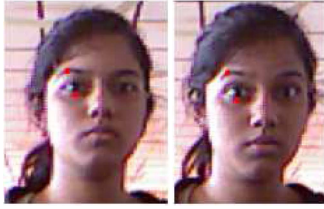


Fig. 13. Emotion – Fear.

The points 16,24 are used to calculate the distance between eyebrows and eye.

We found that when a person is afraid, the distance between the eyebrow and eye is negligible. So we squared to get considerable difference. And we achieved it.

When we coded for the fear detection, it is,

```
if ((abwidth * abwidth) > 260)
{
    tb2.Text = "Fear";
    //Address the fear of the avatar with the key stroke F5.
    keybd_event((byte)Keys.F5, 0x45, KEYEVENTF_EXTENDEDKEY, (UIntPtr)0);
    //This triggers the keyboard event to smile when the user fears.
}
```

Fig. 14. Code snippet for fear

TABLE IV. CONFUSION MATRIX FOR FEAR

	Positive	Negative
True	27	3
False	0	30

E. Anger

When we are angry, we lower our eyebrows. Which means the distance between your eyebrow and eye will be considerably small when compared to normal expression. So, under the given circumstances, when the person is angry, there will be calculate-able difference of distance between eyebrow and eye. Thus, we can detect anger emotion as shown in “Fig. 15”

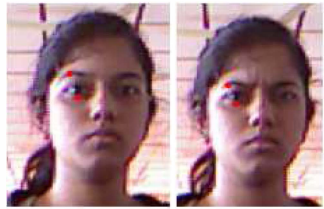


Fig. 15. Emotion – Anger.

The points 16, 24 are used to calculate the distance between eyebrows and eye. We found that when a person is angry, the distance between the eyebrow and eye is negligible. So we squared to get considerable difference. And we achieved it. When we coded for the fear detection, it is,

```
else if ((abwidth * abwidth) < 200)
{
    tb2.Text = "angry";
    //Address the anger of the avatar with the key stroke F6.
    keybd_event((byte)Keys.F6, 0x45, KEYEVENTF_EXTENDEDKEY, (UIntPtr)0);
    //This triggers the keyboard event to be angry when the user is angry.
}
```

Fig. 16. Code snippet for anger

TABLE V. CONFUSION MATRIX FOR ANGER

	Positive	Negative
True	26	4
False	2	28

V. EMOTION DETECTION USING SECOND LIFE

Flexible Action and Articulated Skeleton Toolkit, FFAST, is the middleware to facilitate integration of full-body control with virtual reality applications and video games using Open-NI depth sensor. It is developed by USC for connecting to Second Life.

FFAST works on the phenomenon of tracking the skeleton i.e., users motion using Sensor. FFAST includes a custom VRPN server to stream the user's skeleton over a network, allowing VR applications to read skeletal joints as trackers using any VRPN client.

But, the main drawback in using FFAST in our project is that, it's not capable of tracking the face. Which in-turns find difficult to detect the emotions. So, from the c# program which we used to identify the emotions, we used the ‘Send Keys’ to transfer the keys to the Second Life.

Earlier, we characterized the features of smile, surprise, sad, fear and anger and assigned short cut keys. Later we sent those keys to the second life. Manipulation of code is done in such a way that, we are able to handle with keyboard events.

1. We have added the .dll file which enables for the transfer of the keys

```
...
public partial class MainWindow : Window
{
    [DllImport("user32.dll", SetLastError = true)]
    static extern void keybd_event(byte bVik, byte bScan, uint dwFlags, UIntPtr dwExtraInfo);
}
```

Fig. 17. Code snippet to add ‘.dll’ file.

2. We created a new class, so that all the functions needed are carried out in the class.

3. Invocation of the application is made by calling the instance of the class.

```
int iHandle = NativeWin32.FindWindow(null, "Second Life Viewer");

const int KEYEVENTF_EXTENDEDKEY = 0x1;
const int KEYEVENTF_KEYUP = 0x2;
NativeWin32.SetForegroundWindow(iHandle);
```

Fig. 18. Code snippet for enabling connection.

5. The keyboard strokes are defined where ever, the emotion is detected and trying to connect to second life.

```
keybd_event((byte)Keys.F2, 0x45, KEYEVENTF_EXTENDEDKEY, (UIntPtr)0);
```

Fig. 19. Code snippet for interfacing.

VI. CONCLUSION AND FUTURE WORK

The project has been successfully completed under the Summer Internship Program for the Department of Information Technology, NITK. Various algorithms have been studied and appropriate ones are used. FACS has been used to define the Aus of the emotions. The connection of emotions to the 3D Virtual Avatar of the 3D Virtual world in Second Life is achieved by implementing small changes to the code. Earlier, before taking up this project, there was confusion, if the mesh could identify the features perfectly. Using FACS Algorithm, a conclusion is drawn that, we can track the features even through applying the facial mesh around the person's face. There spotted another difficulty that was of connection to second life. The connection was finally made by adding the '.dll' files to the program which enables the transfer of the commands to Second Life.

The main drawback of the proposed real time emotion recognition system is that disgust emotion is not incorporated. The disgust emotion, when experimented on individuals, shows that it's a combination of anger and sad. Further this system has fundamental limitation that, the emotions can be detected when a person is at a distance of 1meter away from the Kinect depth sensor.

Hence, the future direction of the project is to recognize the emotions of disgust and contempt. This project can be further extended in developing an empathy machine, which

helps the speech and hearing impaired people can easily express their emotions with others.

ACKNOWLEDGMENT

In order to accomplish a task it requires support of many people and well-wishers. Authors would like to express sincere thanks to Mr. Kundan Dere and other members of Department of IT, NITK, Surathkal, Mangalore, India.

First of all I like to express my gratitude to my mentor Prof. G. Ram Mohana Reddy who gave me opportunity and guidance. He gave me useful suggestions that motivated me.

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